

FINITE ELEMENT ANALYSIS

PROFESSIONAL ELECTIVE-II

VI Semester:										
Course Code		Category		Hours / Week			Credits	Maximum Marks		
A6AE33		PEC		L	T	P	C	CIA	SEE	Total
				3	-	-	3	40	60	100
UNIT-I	INTRODUCTION									
Basic concept, general description, application of FEM, , Stress, strain relations, Strain, Displacement relations. Analysis of one dimensional elements, coordinates systems and shape functions. Potential Energy approach: Assembly of Global stiffness matrix and load vector. Finite element equations, boundary conditions.										
UNIT-II	ANALYSIS OF 1D STRUCTURE									
ANALYSIS OF TRUSSES: Stiffness Matrix for plane truss, stress calculations. ANALYSIS OF BEAMS: Hermite shape functions-Element stiffness matrix for two nodes, two degrees of freedom pernode beam element, load vector, deflection, stresses.										
UNIT-III	TWO DIMENSIONAL PROBLEMS									
2-D PROBLEMS: CST-Stiffness matrix and load vector, Isoparametric element representation, Shape functions,convergence requirements, Problems. FINITE ELEMENT MODELLINGof Axisymmetric solids subjected to Axisymmetric loading with triangular elements.Two dimensional four nodedisoparametric elements.										
UNIT-IV	STEADY STATE HEAT TRANSFER ANALYSIS									
STEADY STATE HEAT TRANSFER ANALYSIS: one dimensional analysis of a fin and two dimensional analysis of thin plate.										
UNIT-V	DYNAMIC ANALYSIS									
DYNAMIC ANALYSIS: Formulation of finite element model, element matrices, Lumped and consistent mass matrices-evaluation of Eigen values and Eigen vectors for a stepped bar for free vibrations										
Text Books:										
1. R. Tirupathi Chandrapatla (2011), Introduction to Finite Elements in Engineering, 4 rd edition, Pearson Education, India. 2. V. David. Hutton (2010), Fundamentals of finite elements analysis, 1 st edition, Tata McGraw-Hill education (P) Ltd, New Delhi, India										
Reference Books:										

1. J. N. Reddy (2010), An introduction to Finite Element Method, 3rd edition, Tata McGraw hill education (P) Ltd, New Delhi, India.
2. Chennakesava R. Alavala (2009), Finite elements methods, 1st edition, second reprint, Prentice Hall of India publishers, New Delhi, India.

COURSE OUTCOMES : STUDENTS SHOULD BE ABLE TO

1. Develop elastic equations, formulate and solve the axially loaded bar structures using FEM
2. Apply finite element method to truss and beam analysis.
3. Implement finite element method to solve two dimensional Problems
4. Solve and analyze heat transfer problems using FEM.
5. Formulate the finite element model for the dynamic analysis of one dimensional bar problems